

Shivansh Nigam

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PROFESSIONAL SUMMARY

Final-year engineering undergraduate at MANIT Bhopal with strong foundations in Data Structures & Algorithms and hands-on project experience building full-stack applications, backend APIs, and applied AI/ML solutions. Solved **800+ DSA problems** across competitive programming platforms.

EDUCATION

Maulana Azad National Institute of Technology (MANIT)

Bhopal, MP, India

Bachelor of Technology (B.Tech) in Electrical Engineering

Aug 2023 – May 2027

• **Coursework:** Data Structures & Algorithms, AI/ML, DBMS, Operating Systems, Computer Networks, Linear Algebra.

TECHNICAL SKILLS

Languages & Core CS: C++17, Python, JavaScript (ES6+), TypeScript, SQL, Data Structures & Algorithms, OOP, DBMS, OS

Full-Stack & Backend: Node.js, Express.js, FastAPI, React.js, Next.js, REST APIs, WebSockets, PostgreSQL, Redis, MongoDB

AI/ML & Data Science: Scikit-learn, Pandas, NumPy, SciPy, OpenCV, Explainable AI (OpenXAI), KMeans, Linear Regression

GenAI & Developer Tools: Claude, ChatGPT, Antigravity IDE, Cursor, GitHub Copilot, Gemini/Groq APIs, Docker, Git, Postman

WORK EXPERIENCE

Yasa Ed Tech

Remote

Software Engineering Intern

May 2026 – Jul 2026

- Architected and deployed 15+ RESTful API endpoints and backend services using Node.js, Express.js, and TypeScript, establishing structured data validation and reducing API response latency by **32%**.
- Engineered Redis caching layers and optimized PostgreSQL queries using composite indexing and CTEs, decreasing database load by **40%** and accelerating peak data retrieval times.
- Implemented secure JWT authentication with role-based access control (RBAC) and robust error-handling middleware, ensuring **99.9%** service reliability across core backend pipelines.

TECHNICAL PROJECTS

SleekLink – Enterprise URL Redirection Engine & Analytics Dashboard [GitHub]

Tech Stack: Node.js, Express, PostgreSQL, Redis, React, Vite

- Built a write-through caching layer using Redis over PostgreSQL, achieving **sub-millisecond URL redirection latency**.
- Architected active/pause link state controls with instant Redis cache invalidation, routing paused traffic to custom 403 pages.
- Developed a timezone-aware visitor analytics pipeline with PostgreSQL **AT TIME ZONE** and zero-dependency regex User-Agent parsing.

UPI Scam Detection Bot – Multi-Modal AI Fraud Analysis Engine [GitHub]

Tech Stack: Python, FastAPI, OpenCV, NLP Text Classifier, Llama 3 (Groq API), Redis

- Engineered a multi-modal fraud detection pipeline evaluating SMS text semantics (NLP), phishing URLs, and QR payment payloads (OpenCV).
- Aggregated multi-branch signals into a calibrated 0 – 100 Risk Score across 4 tiers (LOW, MEDIUM, HIGH, CRITICAL) under **150ms**.
- Integrated Llama 3 via Groq API to generate human-readable fraud risk explanations and safety warnings while automatically redacting PII.

Route Navigation Engine – Low-Latency C++ Pathfinding System [GitHub]

Tech Stack: C++17, Node.js, Express, Redis, HTML5 Canvas

- Engineered a C++17 pathfinding core implementing A*, Dijkstra, BFS, and DFS with **1.6 μ s average latency**.
- Architected a hybrid backend combining a Node.js Express API gateway with compiled C++ sub-processes and Redis path caching.
- Built an interactive HTML5 Canvas visualizer supporting dynamic spatial scaling, real-time node selection, and path animations.

AgroMind – AI Decision Support System & Explainable ML Pipeline [GitHub]

Tech Stack: Python, FastAPI, Scikit-learn, Groq API (Llama 3), PostgreSQL, Docker

- Trained a Random Forest classifier on soil chemistry and climate metrics, achieving **94.6% accuracy** for crop yield prediction.
- Built an Explainable AI (XAI) attribution driver computing factor-by-factor feature importance scores to explain model predictions.
- Integrated Llama 3 via Groq API to inject real-time telemetry, weather alerts, and soil metrics into context-aware advisory chats.

HONORS & ACHIEVEMENTS

- Secured **98th Percentile in JEE Main 2023** among 1.2M+ candidates nationwide.
- Solved **800+ Data Structures & Algorithms problems** across competitive platforms (Pupil at Codeforces).

LEADERSHIP & MENTORSHIP

DataBiz (Data Science & Analytics Club, MANIT Bhopal)

Bhopal, MP, India

Technical Lead & Software Mentor

Feb 2025 – Present

- Mentored **100+ students** in full-stack web dev, Python analytics, and SQL; built club web portals and organized campus hackathons.